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Qwen-3D: A Generalist 3D Vision-Language Model for Spatial Understanding

Carnegie Mellon Researchers Unify 2D and 3D Grounding by Computing Attention in World Space Rather Than Image Space

By Lucy Lin, Ayush Jain, Yifan Liu, Katerina Fragkiadaki (Carnegie Mellon University)

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A new generalist vision-language model from Carnegie Mellon University achieves state-of-the-art results on both 2D and 3D spatial understanding benchmarks by a single architectural change: computing attention in 3D world coordinates rather than per-frame pixel coordinates. Qwen-3D surpasses existing 3D LMMs and several large proprietary 2D models on grounding tasks, and will appear at ECCV 2026.

AT A GLANCE

Problem: Frame-centric tokenization ignores the shared 3D structure underlying multi-view observations, limiting VLMs on long videos and 3D scenes.

Why it matters: Embodied AI and robotics need a single model that reasons about 3D space and natural language without separate specialist pipelines.

Method: 3D Rotary Positional Embeddings index attention by world-space coordinates; a unified output head predicts language, 2D boxes, and 3D boxes.

Result: State-of-the-art on ScanQA, EmbodiedScan, and other 3D benchmarks; competitive with large proprietary 2D models on MMBench and SEED-Bench.

Evidence: Evaluated across six 3D understanding and five 2D vision-language benchmarks at ECCV 2026.

The Big Picture

Large Multimodal Models have transformed image and short-video understanding, but scaling to 3D scenes and long videos has remained elusive. Every RGB frame is tokenized independently, discarding the geometric relationships that tie views of the same scene together. A model watching a robot navigate a room from multiple cameras sees a stream of disconnected image patches, with no built-in sense that the sofa seen from Camera 1 is the same object partially visible in Camera 2.

Qwen-3D solves this by lifting attention from 2D pixel

space into 3D world space. Given depth maps and camera poses alongside RGB images, the model maps every visual token to a 3D world coordinate and uses that coordinate as the token’s positional encoding. Tokens that correspond to the same 3D point—even from different cameras or timesteps—receive encodings that reflect their world-space proximity, enabling the model to naturally associate them during attention without any explicit correspondence algorithm.

Why Existing Approaches Fall Short

Prior 3D LMMs (e.g. 3D-VisTA, EmbodiedScan) added 3D-awareness through auxiliary encoders or separate perception heads, but kept the core transformer operating in 2D image space. This created a mismatch: the language backbone reasons over 2D feature sequences while 3D grounding heads operate over a separate 3D feature space, requiring careful multi-task balancing and making joint 2D/3D generalization difficult.

Specialist 3D perception models (Mask3D, OpenMask3D) achieve strong grounding results but are not language models and cannot engage in open-ended dialogue or instruction following. Generalist VLMs (GPT-4V, Gemini Pro Vision) handle language fluently but treat 3D scenes as collections of independent image crops, losing spatial coherence. Qwen-3D is the first model to close all three gaps in a single architecture.

Core Mechanism

3D Token Lifting. For each pixel (u, v) in an input frame, depth d and camera intrinsics K and extrinsics $[R|t]$ yield the world coordinate:

$$\mathbf{p} = R^{-1} \left(d K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} - t \right).$$

Each visual token (a patch of pixels) is assigned the 3D centroid of its patch.

3D Rotary Positional Embeddings (3D RoPE). Standard RoPE encodes 1D or 2D positions by rotating query and key vectors in frequency-indexed subspaces. Qwen-3D extends this to 3D: the world coordinates (X, Y, Z) are used to construct rotation matrices for three independent groups of attention head dimensions, one per spatial axis. Two tokens close in 3D world space receive nearly identical positional rotations, producing high raw dot-product similarity before any content-based weighting.

Unified Output Head. A single transformer decoder produces three output types from the same representation: free-form text, 2D bounding boxes (normalized image coordinates), and 3D bounding boxes (centre, size, heading angle in world coordinates). Task type is controlled by

instruction prompts, enabling zero-shot switching between modalities.

Technical Formulation

Let $\mathbf{F} = \{(\mathbf{p}_i, \mathbf{v}_i)\}_{i=1}^N$ be the set of N visual tokens with 3D world positions $\mathbf{p}_i \in \mathbb{R}^3$ and feature vectors $\mathbf{v}_i \in \mathbb{R}^d$. Self-attention with 3D RoPE applies:

$$\text{Attn}(Q, K, V)_{ij} = \frac{(R(\mathbf{p}_i) q_i)^\top (R(\mathbf{p}_j) k_j)}{\sqrt{d_h}},$$

where $R(\mathbf{p}) \in \mathbb{R}^{d_h \times d_h}$ is the block-diagonal rotation matrix constructed from the three spatial frequencies of $\mathbf{p}$.

The training objective combines cross-entropy for text tokens ($\mathcal{L}_{\text{LM}}$), smooth- ℓ_1 for 2D box coordinates ($\mathcal{L}_{2\text{D}}$), and smooth- ℓ_1 plus IoU3D loss for 3D box prediction ($\mathcal{L}_{3\text{D}}$):

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_1 \mathcal{L}_{2\text{D}} + \lambda_2 \mathcal{L}_{3\text{D}}.$$

Learning or Inference Procedure

Data. Training uses a mixture of (i) standard 2D VQA and instruction-following datasets, (ii) multi-view indoor scene datasets with 3D annotations (ScanRefer, Nr3D), and (iii) egocentric video datasets with depth and camera poses.

Fine-tuning. Starting from a Qwen-VL checkpoint, the model is fine-tuned end-to-end with the multi-task objective above. The 3D RoPE replaces the existing 2D RoPE positional encoding; no architectural changes to the attention mechanism itself are required.

Inference. At test time, the model accepts any combination of RGB images, depth maps, and camera poses. Depth-free inference falls back to 2D RoPE, preserving 2D benchmark performance even when 3D information is unavailable.

What the Guarantee Says

3D RoPE preserves the rotation equivariance of standard RoPE: if all tokens are rigidly rotated in world space (equivalent to a global camera pose change), the relative positional encoding between any pair of tokens is unchanged, making the model invariant to global reference frame choice. The unified training objective ensures that improving 3D grounding does not degrade language modeling performance—empirically confirmed by stable scores on 2D benchmarks throughout training.

Experimental Findings

- **ScanQA:** new state of the art, surpassing previous best 3D LMM by a significant margin on both EM and CIDEr metrics.

- **EmbodiedScan:** top result on 3D visual grounding (Acc@0.25 and Acc@0.5), outperforming specialist 3D perception pipelines.
- **MMBench / SEED-Bench:** competitive with GPT-4V and Gemini Pro Vision on standard 2D multimodal benchmarks, demonstrating no 3D/2D trade-off.
- **Multi-view video understanding:** large improvements on tasks requiring cross-view association (e.g. object re-identification across camera angles).

Ablations and Interpretation

3D RoPE vs. 2D RoPE. Removing 3D positional information and reverting to per-frame 2D RoPE drops 3D grounding accuracy by over 15 points on ScanRefer, confirming that world-space position is the key ingredient rather than multi-view data alone.

Depth quality sensitivity. Using predicted depth (from a monocular estimator) instead of ground-truth depth degrades 3D grounding by 4–6 points but still outperforms prior 3D LMMs that use GT depth, suggesting the 3D positional signal is robust to moderate depth noise.

Joint vs. separate 2D/3D training. Training only on 3D data degrades 2D benchmark performance by 8 points; joint training recovers it fully, confirming that the two modalities are complementary rather than competing.

Reference

Lucy Lin, Ayush Jain, Yifan Liu, Katerina Fragkiadaki. **Qwen-3D: A Generalist 3D Vision-Language Model for Spatial Understanding.** arXiv:2608.02980, ECCV 2026. <https://arxiv.org/abs/2608.02980>

Keywords: post-training quantization • multi-precision LLM
• residual quantization • efficient inference

Recurrent Residual Quantization: A Progressive Multi-Precision Representation for LLMs

One Checkpoint, Four Bit-Widths: A Calibration-Free Framework that Builds 2-, 4-, 6-, and 8-Bit Models from a Single Residual Stack

By Yu Luo, Bo Dong, Wenhua Cheng, Haihao Shen

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Deploying LLMs across heterogeneous hardware—from cloud GPUs to edge devices—today requires a separate quantized checkpoint for every target bit-width. Recurrent Residual Quantization (RRQ) eliminates this overhead by representing weights as a base plus a sequence of 2-bit residual corrections: reading only the base gives 2-bit inference; reading base plus k residuals gives $(2 + 2k)$ -bit inference. The full 2/4/6/8-bit package for Qwen3-8B is constructed in 1,293 seconds, $3.3\times$ faster than MatGPTQ, with no calibration data required.

AT A GLANCE

Problem: Each target bit-width requires a separate quantization run and checkpoint, multiplying storage and maintenance costs linearly with the number of deployment targets.

Why it matters: A single multi-precision checkpoint enables hardware-adaptive serving: the same model file serves data-center GPUs at 8-bit and edge CPUs at 2-bit.

Method: Progressive 2-bit residuals applied to the quantization error of the previous step; no calibration data; no joint multi-bit optimization.

Result: $3.3\times$ faster construction than MatGPTQ; competitive perplexity at all four precisions; single-file deployment.

Evidence: Benchmarked on Qwen3-8B and LLaMA-3-8B; accepted to NeurIPS 2026.

The Big Picture

Large language models are increasingly deployed across a spectrum of hardware: high-memory data-center GPUs running 8- or 16-bit inference, on-device accelerators requiring 4-bit, and low-power edge CPUs that may only support 2-bit integer arithmetic. Current practice requires a separate post-training quantization (PTQ) run—often

including an expensive calibration step—for each target precision. A team deploying Llama-3 at 2-, 4-, and 8-bit must run three separate GPTQ pipelines and store three separate checkpoints.

Recurrent Residual Quantization (RRQ) collapses this into one. A single RRQ construction run produces a structured checkpoint from which any of the four target precisions can be read at inference time by selecting how many correction layers to load. There is no trade-off in quality: each precision performs comparably to a dedicated GPTQ checkpoint optimized for that bit-width.

Why Existing Approaches Fall Short

Calibration-based PTQ (GPTQ, AWQ, MatG-PTQ) achieves near-FP16 quality at 4-bit and above, but requires a calibration dataset (typically 128 sequences of 2048 tokens) and second-order Hessian computations that dominate construction time. A new bit-width requires a new run; the runs cannot be composed.

Residual Vector Quantization (RVQ) decomposes weights into a codebook plus residual codebooks, enabling multiple precisions. However, RVQ optimizes all codebooks jointly, which is expensive, and the discrete codebook structure is poorly suited to the per-layer, per-group statistics of LLM weights.

Non-uniform quantization (NF4, SpQR) adapts bit allocation to weight distributions but is fixed to a single precision per run.

RRQ’s key insight is that greedy round-to-nearest residuals—not joint optimization—are sufficient to close the quality gap between adjacent bit-widths, because each residual corrects for the error of the previous quantization step independently.

Core Mechanism

RRQ builds a weight approximation $\hat{w}^{(k)}$ by iteratively quantizing the error of the previous approximation:

$$\hat{w}^{(0)} = Q_2(w), \quad r^{(j)} = w - \hat{w}^{(j-1)}, \quad \hat{w}^{(j)} = \hat{w}^{(j-1)} + Q_2(r^{(j)}),$$

where $Q_2(\cdot)$ denotes round-to-nearest 2-bit scalar quantization with per-group scales computed from the range of the argument. After k iterations, the approximation has effective precision $(2 + 2k)$ bits. Concretely:

- $\hat{w}^{(0)}$: effective 2-bit (base only)
- $\hat{w}^{(1)}$: effective 4-bit (base + 1 residual)
- $\hat{w}^{(2)}$: effective 6-bit (base + 2 residuals)
- $\hat{w}^{(3)}$: effective 8-bit (base + 3 residuals)

The checkpoint stores $\hat{w}^{(0)}$ and $r^{(1)}, r^{(2)}, r^{(3)}$ as separate tensors; inference at precision $(2 + 2k)$ loads and sums $\hat{w}^{(0)}$ through $Q_2(r^{(k)})$.

Technical Formulation

Let $W \in \mathbb{R}^{m \times n}$ be a weight matrix partitioned into groups of size g . For each group $\mathcal{G}_i$, the 2-bit quantizer is:

$$Q_2(x; \mathcal{G}_i) = \text{round}\left(\frac{x - \min(\mathcal{G}_i)}{\max(\mathcal{G}_i) - \min(\mathcal{G}_i)} \cdot 3\right) \cdot \frac{\max(\mathcal{G}_i) - \min(\mathcal{G}_i)}{3} + \min(\mathcal{G}_i)$$

The reconstruction error after k residual layers satisfies:

$$\|w - \hat{w}^{(k)}\|_2^2 = \|r^{(k)} - Q_2(r^{(k)})\|_2^2 \leq \frac{(\Delta^{(k)})^2}{4},$$

where $\Delta^{(k)} = \max(r^{(k)}) - \min(r^{(k)})$ is the range of the k -th residual, which shrinks geometrically with k as the residuals concentrate near zero.

Learning or Inference Procedure

Construction. For each linear layer, the RRQ construction procedure: (1) computes the 2-bit base $\hat{w}^{(0)}$ via RTN with per-group scales; (2) computes residuals $r^{(j)}$ and their 2-bit quantizations sequentially; (3) stores the base and all residuals. No calibration data is touched; construction is embarrassingly parallel across layers.

Inference. A deployment configuration specifies the target precision $b \in \{2, 4, 6, 8\}$. The runtime loads the base and $(b - 2)/2$ residual tensors, sums them, and uses the result as the weight matrix. Memory footprint is proportional to b ; switching precision at inference time requires only loading more or fewer residual tensors, not re-quantizing.

Hardware kernels. Because both the base and each residual are 2-bit integer tensors with per-group scales, existing 2-bit matmul kernels (e.g. INT2 CUDA kernels) apply directly. The multi-residual accumulation adds one extra addition per residual per layer.

What the Guarantee Says

The greedy residual construction gives a monotone quality improvement: adding each residual layer can only reduce or maintain the reconstruction error, never increase it. Formally, $\|w - \hat{w}^{(k)}\|_2 \leq \|w - \hat{w}^{(k-1)}\|_2$ for all k . This ensures that the 4-bit model is always at least as good as the 2-bit model, and the 6-bit model at least as good as the 4-bit model, from the same checkpoint.

Experimental Findings

- **Construction time (Qwen3-8B):** 1,293 seconds for the full 2/4/6/8-bit package, 3.3× faster than MatGPTQ (4,267 seconds).
- **Perplexity (WikiText-2):** RRQ-4bit matches GPTQ-4bit within 0.1 perplexity points on Qwen3-8B; RRQ-8bit is within 0.05 of FP16.

- **Zero-shot accuracy (MMLU, HellaSwag):** <0.5 point degradation at 4-bit vs. FP16 baseline on both Qwen3-8B and LLaMA-3-8B.
- **Storage:** The complete 4-precision package is 1.15× the size of a single 8-bit GPTQ checkpoint.

Ablations and Interpretation

Group size. Larger groups (128 vs. 64) reduce storage but increase 2-bit error; the residual correction partially compensates, narrowing the quality gap between group sizes. At 4-bit and above, group size has negligible effect.

RTN vs. GPTQ base. Starting from a GPTQ-optimized 2-bit base (rather than RTN) yields 0.3–0.5 perplexity improvement at 2-bit but does not meaningfully affect 4-bit or higher precision, confirming that the residuals correct for RTN’s error.

Number of residuals. Quality saturates after 3 residuals (8-bit effective), matching the known empirical ceiling of scalar PTQ beyond 8-bit for LLMs.

Reference

Yu Luo, Bo Dong, Wenhua Cheng, Haihao Shen. **Recurrent Residual Quantization: A Progressive Multi-Precision Representation for LLMs.** arXiv:2608.04048, NeurIPS 2026. <https://arxiv.org/abs/2608.04048>

Keywords: offline reinforcement learning • diffusion policy • Q-value correction • distribution shift

Diffusion Policy with Behavioral Advantage Correction for Offline Reinforcement Learning

BAC-PE Uses the Behavior Policy’s Q-Function to Simultaneously Tame Overestimation and Conservatism in Offline RL

By Botao Dong, Longyang Huang, Ning Pang, Hongtian Chen

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Offline reinforcement learning is caught between two failure modes: over-conservative penalization of in-distribution actions and overestimation of out-of-distribution actions. Behavioral Advantage Corrected Policy Evaluation (BAC-PE) resolves this dilemma by adding the behavior policy’s advantage function as a correction term to the learned Q-function, with convergence guarantees and strong empirical results across D4RL, Antmaze, and robotic manipulation benchmarks.

AT A GLANCE

Problem: Distribution shift in offline RL causes Q-functions to either overestimate out-of-distribution actions or over-penalize in-distribution ones.

Why it matters: Safe offline RL from logged data is essential for robotics and healthcare, where online interaction is expensive or dangerous.

Method: Correct the learned Q-function with the behavior policy’s advantage; represent both policies as diffusion models for multimodal action distributions.

Result: Outperforms CQL, TD3+BC, and Diffusion-QL on D4RL locomotion and Antmaze; strong results on robotic manipulation.

Evidence: Theoretical convergence bound on Q-estimation error; evaluated on D4RL, Antmaze, and Robosuite benchmarks.

The Big Picture

Offline reinforcement learning promises to extract useful policies from logs of past behavior—a critical capability for domains such as healthcare and autonomous driving where online interaction is too costly or dangerous. The fundamental challenge is that the learned policy will inevitably visit state-action pairs not well represented in the fixed dataset. Standard Q-learning then propagates erroneous value estimates for these out-of-distribution (OOD)

actions, causing the policy to exploit them catastrophically.

The field has responded with two families of corrections. *Conservative* methods (CQL, IQL) penalize OOD actions with explicit regularization, but the penalty is necessarily blunt: it can suppress well-covered in-distribution actions alongside genuinely dangerous OOD ones. *Implicit constraint* methods (TD3+BC, BEAR) restrict the policy to stay close to the behavior distribution, but tight constraints sacrifice the ability to improve over a mediocre behavior policy. BAC-PE takes a fundamentally different approach: rather than penalizing OOD actions by their distance from the behavior distribution, it *corrects* Q-values using information the behavior policy actually provides.

Why Existing Approaches Fall Short

The behavior policy μ generates all data in the offline dataset. Its Q-function Q^μ is estimable from the data without distribution shift, because every action in the dataset was taken by μ on the state distribution that μ itself induced. This is the key observation that existing methods leave on the table.

Conservative Q-Learning (CQL) adds a penalty $-\mathbb{E}_{a \sim \pi}[Q]$ to push Q-values down for policy actions and up for data actions. But it does so uniformly across all policy actions, regardless of whether they are genuinely OOD or merely low-density actions that μ took rarely but which are still valid.

TD3+BC adds a behavioral cloning regularizer $\|a - a_\mu\|^2$ to the policy gradient, preventing large deviations from the logged actions but also preventing the policy from taking shortcuts the behavior policy missed.

Neither method uses Q^μ directly. BAC-PE does.

Core Mechanism

Behavioral Advantage. The behavioral advantage of action a in state s is:

$$A^\mu(s, a) = Q^\mu(s, a) - V^\mu(s),$$

where $V^\mu(s) = \mathbb{E}_{a \sim \mu}[Q^\mu(s, a)]$. A positive behavioral advantage means a is *better than average* under μ ; a negative behavioral advantage means a is OOD or poor.

BAC-PE Correction. The corrected Q-function for the learned policy π is:

$$\hat{Q}^\pi(s, a) = Q^\pi(s, a) + \alpha A^\mu(s, a).$$

The intuition:

- If a was frequently taken and good under μ (large $A^\mu > 0$), $\hat{Q}^\pi$ is boosted, offsetting excessive conservatism.

- If a was rare or poor under μ (small or negative A^μ), $\hat{Q}^\pi$ is suppressed, offsetting overestimation.

No uniform penalty is applied; the correction is adaptive, action-by-action.

Diffusion Policy. Both μ and π are represented as *diffusion models* over the action space:

$$\pi_\theta(a|s) \propto \int p_{\theta,T}(a_T) \prod_{t=1}^T p_{\theta,t}(a_{t-1}|a_t, s) da_{1:T},$$

where $a_T \sim \mathcal{N}(0, I)$ and $p_{\theta,t}$ is a learned denoising step. Diffusion policies capture multimodal action distributions (e.g. multiple valid grasps for a manipulation task) that Gaussian or deterministic policies cannot.

Technical Formulation

Let $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}$ be the offline dataset collected under μ . BAC-PE estimates Q^μ and V^μ via standard TD with dataset transitions:

$$Q_\phi^\mu \leftarrow \arg \min_{\phi} \mathbb{E}_{\mathcal{D}} \left[\left(Q_\phi^\mu(s, a) - r - \gamma V_\phi^\mu(s') \right)^2 \right],$$

then learns Q^π with the corrected target:

$$\hat{Q}_\psi^\pi \leftarrow \arg \min_{\psi} \mathbb{E}_{\mathcal{D}} \left[\left(\hat{Q}_\psi^\pi(s, a) - r - \gamma \hat{Q}_\psi^\pi(s', a') \right)^2 \right],$$

where $a'_{s'} \sim \pi_\theta(\cdot|s')$.

The policy is optimized to maximize $\hat{Q}^\pi$ subject to a KL constraint on deviation from μ :

$$\mathcal{L}(\theta) = -\mathbb{E}_{s,a \sim \pi_\theta} \left[\hat{Q}^\pi(s, a) \right] + \lambda D_{\text{KL}}(\pi_\theta(\cdot|s) \parallel \mu(\cdot|s)).$$

Learning or Inference Procedure

Stage 1: Behavior policy fitting. A diffusion model is fit to the offline dataset actions conditioned on states: this is standard behavioral cloning with a diffusion denoiser.

Stage 2: Q-function estimation. Q_ϕ^μ and V_ϕ^μ are trained via TD using the offline dataset.

Stage 3: Policy optimization. The diffusion policy π_θ is optimized alternately updating the denoising network (to maximize $\hat{Q}^\pi$) and the Q-function Q_ψ^π (to provide targets for the next policy update).

Inference. At test time, the diffusion policy generates actions by running T denoising steps from Gaussian noise conditioned on the current state.

What the Guarantee Says

Theorem (informal). Under standard concentrability and function approximation assumptions, the BAC-PE Q-function Q_k^π at iteration k satisfies:

$$\|Q_k^\pi - Q^{\pi^*}\|_\infty \leq \varepsilon_k + C \delta_\mu,$$

where $\varepsilon_k \rightarrow 0$ as $k \rightarrow \infty$, C is a problem constant, and $\delta_\mu = \sup_{s,a} (\pi(a|s)/\mu(a|s))$ is the concentrability coefficient. Actions well-covered by μ (δ_μ small) converge to their true Q-values; error grows gracefully as actions move OOD.

Experimental Findings

- **D4RL locomotion** (HalfCheetah, Hopper, Walker2D, Ant): BAC-PE improves over Diffusion-QL by 3–7 normalized score points on medium-expert datasets, the most challenging offline distribution.
- **Antmaze** (medium-play, large-diverse): 8–12 success rate improvement over CQL, which is excessively conservative on long-horizon tasks.
- **Robosuite manipulation** (Lift, PickPlace): 15% improvement in task completion over TD3+BC, where multimodal action distributions are critical.

Ablations and Interpretation

Advantage coefficient α . Setting $\alpha = 0$ (no correction) gives standard diffusion Q-learning; increasing α improves performance up to $\alpha \approx 1$ and then degrades, consistent with the theoretical bound: too large α overweights behavior-policy information at the expense of the true Q-values.

Diffusion vs. Gaussian policy. Replacing the diffusion policy with a Gaussian policy degrades performance by 5–8 points on manipulation tasks where action distributions are strongly multimodal, confirming that representation quality matters alongside the BAC-PE correction.

Q^μ estimation quality. Using a poorly estimated Q^μ (trained on a small data subset) degrades performance, confirming that BAC-PE’s advantage is contingent on reliable behavior-policy value estimation.

Reference

Botao Dong, Longyang Huang, Ning Pang, Hongtian Chen. **Diffusion Policy with Behavioral Advantage Correction for Offline Reinforcement Learning.** arXiv:2608.02332, August 2026. <https://arxiv.org/abs/2608.02332>

Keywords: self-consistency • sycophancy • LLM training objectives • cross-input consistency

Position: It's Time to Optimize LLMs for Self-Consistency

MIT Researchers Argue that Sycophancy, Logical Incompleteness, and Overconfidence Share a Single Root Cause—and a Single Fix

By Itamar Pres, Belinda Z. Li, Laura Ruis, Zifan Carl Guo, Keya Hu, Mehul Damani, Isha Puri, Ekdeep Singh Lubana, Jacob Andreas (MIT CSAIL et al.)

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Sycophancy, overconfident wrong answers, and incomplete logical generalization are three of the most persistent failures in large language models. This position paper argues they share a single root cause—point-wise training that evaluates behavior on individual input-output pairs—and a single remedy: explicit self-consistency optimization as a first-class training objective.

AT A GLANCE

Problem: LLM training treats behavior as evaluable on single (input, output) pairs, making it blind to inconsistencies across inputs that are logically related.

Why it matters: In high-stakes domains (medicine, law), inconsistency is a safety failure regardless of average per-question accuracy.

Method: Formalize consistency optimization as a cross-input objective; show that paraphrase robustness, logical regularization, and calibration are special cases.

Result: A unified framework that exposes three major failure modes as one problem and proposes consistency benchmarks and training procedures.

Evidence: Theoretical unification; empirical survey of consistency failures across model families and benchmark types.

The Big Picture

Point-wise loss functions—cross-entropy on individual tokens, pairwise preference from RLHF—have powered the dramatic improvements of modern LLMs. But they share a structural blind spot: they evaluate each (prompt, response) pair independently, with no signal about how the model’s behavior should relate across different prompts.

A model that correctly answers “Is the Earth flat?” with “No” and also agrees with a user who insists “But surely

the Earth is flat?” is simultaneously right and wrong on two instances. A standard per-question evaluation would record one correct answer and one incorrect answer. A consistency evaluation records something more damning: the model has no stable beliefs, only context-conditioned agreement.

This paper argues that a wide class of LLM failures—not just sycophancy but logical incompleteness and overconfidence—are manifestations of the same underlying gap: the model has been optimized to produce good-looking individual outputs without any constraint that those outputs relate to each other coherently.

Why Existing Approaches Fall Short

Larger models. Scaling alone does not fix sycophancy; larger models are arguably more sycophantic because they are better at detecting and accommodating user intent from subtle cues in phrasing.

RLHF with human feedback. Human annotators judge responses in context, which can itself introduce sycophancy into the preference labels. Annotators tend to prefer responses that agree with their priors, so training on their preferences reinforces agreement over correctness.

Targeted patches. Adding paraphrase robustness data, logical consistency regularizers, or calibration objectives addresses each failure mode individually but does not generalize: a model patched for paraphrase inconsistency may still be sycophantic, because neither intervention targets the shared root cause.

Core Mechanism

The authors identify a relation structure on inputs:

$$\mathcal{R} = \{(x, x') : f(x) \text{ and } f(x') \text{ should be consistent}\},$$

where “consistent” is defined task-specifically:

- **Paraphrase pairs:** (x, x') are semantic paraphrases; $f(x) = f(x')$ for factual questions.
- **Logical pairs:** x' is a logical consequence of x ; the model asserting x entails asserting x' .
- **Pressure pairs:** x' is x with a sycophantic framing (e.g. user expresses disagreement); $f(x) = f(x')$ for factual questions.
- **Confidence pairs:** a set of instances with the same ground truth; stated confidence should match empirical accuracy.

The consistency loss is:

$$\mathcal{L}_{\text{cons}}(\theta) = \mathbb{E}_{(x, x') \sim \mathcal{R}}[d(f_{\theta}(x), f_{\theta}(x'))],$$

where d is an appropriate divergence (KL for distributions, ℓ_2 for scalar confidence, 0/1 disagreement for discrete

answers). The full training objective is:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{point}}(\theta) + \lambda \mathcal{L}_{\text{cons}}(\theta).$$

Technical Formulation

The paper formalizes the unification via a common optimization template. Let $\{x, x'\} \in \mathcal{R}$ be a consistency pair. Define:

$$\text{Inconsistency}(\theta; x, x') = d(f_\theta(x), f_\theta(x')).$$

Three established methods are special cases:

Paraphrase robustness (e.g. SelfCheckGPT): $\mathcal{R}$ = paraphrase pairs, d = discrete disagreement, f = argmax answer.

LoGIC (logical consistency): $\mathcal{R}$ = logical entailment pairs, $d = \mathbb{I}[\text{contradicts}]$.

Calibration (e.g. RLCF): $\mathcal{R}$ = accuracy–confidence pairs over question sets, d = squared deviation between stated probability and empirical correctness rate.

The consistency framework makes the shared structure explicit and enables joint optimization over all three relation types simultaneously.

Learning or Inference Procedure

Consistency pair construction. For each training example x , generate consistency-required counterparts x' :

- Paraphrase: use a paraphrase model or template augmentation.
- Logical: generate entailed statements via template rules or a symbolic reasoner.
- Pressure: append sycophantic suffixes (“But actually I think the answer is [wrong answer]. Do you agree?”) to factual questions.

Training. Include $\mathcal{L}_{\text{cons}}$ as an additional loss term during SFT or RLHF. The consistency pairs require forward passes on both x and x' , doubling the batch computation per pair but not changing the training algorithm.

Evaluation. Consistency benchmarks evaluate pairs rather than individual instances: for each factual question, measure the frequency with which the model changes its answer under paraphrase, pressure, or logical extension.

What the Guarantee Says

The formal claim is that consistency optimization provides a sufficient condition for avoiding sycophancy, overconfidence, and logical incompleteness *simultaneously*. Specifically: if $\mathcal{L}_{\text{cons}}(\theta) = 0$ over the full relation $\mathcal{R}$ covering all three pair types, then the model exhibits none of the three failure modes. This is a theoretical desideratum that motivates the unified approach; achieving $\mathcal{L}_{\text{cons}} = 0$ in

practice requires the appropriate relations to be covered by training data.

Experimental Findings

- Across GPT-4, Gemini, and Llama families, paraphrase inconsistency rates range from 12% to 31% on factual QA—a large gap invisible to per-question accuracy.
- Sycophancy (answer change under user pressure) occurs in 18–42% of cases across models, increasing with model scale.
- Logical incompleteness: models that state a rule correctly apply it to novel instances only 60–80% of the time.
- Models fine-tuned with consistency objectives show 40–60% reductions in inconsistency rates while maintaining or improving per-question accuracy.

Ablations and Interpretation

Which relation type matters most? Pressure pairs (sycophancy) produce the largest absolute consistency improvement when included in training, likely because sycophantic RLHF feedback has been the dominant training signal for recent models.

λ sensitivity. Setting λ too high degrades per-question accuracy, suggesting a trade-off between point-wise and consistency optimization that requires tuning. The authors recommend $\lambda \in [0.1, 0.5]$ as a safe range.

Generalization to unseen pair types. Models trained with paraphrase consistency also show reduced sycophancy, but less than models trained with pressure pairs directly, suggesting partial but incomplete transfer between consistency pair types.

Reference

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Mamba with Hierarchical Memory: Solving the Representation Bottleneck in Long Sequence Modeling

HMM Adds Paragraph-Level Semantics and Persistent Long-Term Memory to Mamba’s Fixed Recurrent State, Enabling Near-Linear Long-Context Reasoning

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State space models like Mamba offer linear-time sequence processing, but their fixed-capacity recurrent state is a fundamental bottleneck on long sequences: old context is progressively overwritten as the sequence grows. Hierarchical Memory Mamba (HMM) resolves this by adding two lightweight memory tiers above the base recurrent state—paragraph-level semantic extraction and persistent long-term memory retrieval—enabling near-linear complexity long-range reasoning on sequences where Mamba’s performance degrades.

AT A GLANCE

Problem: Mamba’s fixed-size recurrent state overwrites old context as sequences grow, creating a representation bottleneck that limits long-range recall.

Why it matters: Long-document QA, many-shot ICL, and genomic sequence modeling require retaining information across scales far beyond the recurrent state capacity.

Method: Hierarchical Memory Mamba (HMM) extracts paragraph-level semantics from the hidden state and stores them in a persistent long-term memory for task-relevant retrieval.

Result: Large improvements on Long-Range Arena, SCROLLS benchmarks, and many-shot ICL tasks; near-linear complexity vs. quadratic for Transformers.

Evidence: Evaluated on six Long-Range Arena tasks, four SCROLLS benchmarks, and PG-19 language modeling up to 32k tokens.

The Big Picture

Recurrent sequence models like Mamba process tokens in $O(N)$ time and $O(1)$ memory per step, making them far cheaper than Transformers for long sequences. But

the constant memory comes with a hard limit: the recurrent hidden state $h_t \in \mathbb{R}^d$ must compress *all* context seen so far into a fixed-size vector at every step. On short sequences, this works well. On sequences of thousands or tens of thousands of tokens, earlier context is progressively overwritten—a phenomenon the authors call the *representation bottleneck*.

The problem is architectural, not a matter of training. No amount of fine-tuning on long sequences fixes the fundamental capacity limit of a fixed recurrent state. HMM attacks the bottleneck directly by introducing a structured memory hierarchy above the recurrent state, inspired by the cognitive science of human memory (sensory $\rightarrow$ working $\rightarrow$ long-term memory).

Why Existing Approaches Fall Short

Larger recurrent state. Expanding d reduces the bottleneck but increases per-step compute quadratically and does not scale gracefully.

Mamba-2 / HGRN2. These architectural improvements to SSMs increase state capacity and improve practical long-context performance, but the fixed-state bottleneck remains—performance still degrades on Path-X and other tasks requiring recall beyond the effective state horizon.

Hybrid Mamba-Transformer models (Jamba, Zamba) insert full attention layers at regular intervals, recovering long-range recall at the cost of $O(N^2)$ for those layers. HMM achieves near-linear complexity throughout by using sparse key-value retrieval rather than full attention for the long-term memory.

KV-cache SSMs. Some recent models cache a rolling window of past hidden states and attend over them (e.g. RWKV-7’s chunk attention). Window-limited KV caches still impose a horizon; HMM’s long-term memory has no fixed horizon.

Core Mechanism

HMM adds two memory tiers to a standard Mamba backbone:

Tier 1—Sensory Memory (Mamba hidden state). The standard recurrent state h_t processes the current token in the context of recent tokens. It is fast and fine-grained but has fixed capacity d .

Tier 2—Working Memory (PLS extractor). Every P tokens (a “paragraph”), a lightweight cross-attention module reads the P hidden states $h_{[kP:(k+1)P]}$ and compresses them into a single *Paragraph-Level Semantic* (PLS) vector:

$$\text{PLS}_k = \text{MultiHead}(Q_{\text{probe}}, K = W_K H_k, V = W_V H_k),$$
 where $H_k = [h_{kP}, \dots, h_{(k+1)P-1}] \in \mathbb{R}^{P \times d}$ and $Q_{\text{probe}} \in$

$\mathbb{R}^{q \times d}$ is a learned query matrix ($q \ll P$). The PLS captures the slow, paragraph-scale semantics that the fast sensory memory cannot retain.

Tier 3—Long-Term Memory (LTM store). PLS vectors accumulate in a growing bank $\mathcal{M} = \{(\text{PLS}_k, k)\}$. At each step, the current hidden state queries the bank via k -NN retrieval followed by a small attention:

$$\text{retrieved}_t = \text{Attn}(W_q h_t, K_{\text{LTM}}, V_{\text{LTM}}),$$

and the retrieved content is injected into the Mamba state update via a gating mechanism:

$$h'_t = h_t + \sigma(W_g h_t) \odot \text{retrieved}_t.$$

Technical Formulation

The complexity of HMM for a sequence of N tokens with paragraph size P :

Sensory processing: $O(Nd)$ (linear, standard Mamba).

PLS extraction: One cross-attention over P tokens per paragraph: $O((N/P) \cdot P \cdot q \cdot d) = O(Nqd/1)$, effectively $O(N)$ for fixed q .

LTM retrieval: N/P entries in the bank; each retrieval is $O(N/P)$ for exact attention over all entries, giving $O(N \cdot N/P) = O(N^2/P)$ total. For $P \gg 1$ (e.g. $P = 256$), this is orders of magnitude cheaper than full attention ($O(N^2)$).

The training objective adds no new loss terms: HMM is trained end-to-end on the next-token prediction objective, with the PLS extractor and LTM read/write operations differentiable throughout.

Learning or Inference Procedure

Initialization. HMM can be initialized from a pretrained Mamba checkpoint; only the PLS extractor and LTM projection matrices are new parameters (typically $<5\%$ of total parameters).

Fine-tuning. Continued pre-training on long-document corpora for 10–20k steps suffices to activate the LTM pathway; full long-context fine-tuning with HMM from scratch is also supported.

Inference. The LTM bank grows as the sequence is processed; retrieval uses approximate nearest neighbors (FAISS) for bank sizes above 10k entries. The k -NN search cost is sub-linear in practice due to FAISS’s IVF indexing.

What the Guarantee Says

Monotone recall. A PLS vector is written for every paragraph in the input without overwriting; the LTM bank is append-only. This guarantees that information from paragraph k is always retrievable from step $k' > k$, unlike the sensory state which is overwritten at step $k + 1$.

Complexity bound. For paragraph size $P = O(\sqrt{N})$, LTM retrieval cost is $O(N^{3/2})$ —between linear and quadratic. For fixed P , it is $O(N^2/P)$, which for $P = 256$ and $N = 32,768$ is $128\times$ cheaper than full attention.

Experimental Findings

- **Long-Range Arena:** HMM outperforms Mamba-2 by 4–9 points on Path-X and Retrieval tasks, where long-range recall is decisive.
- **SCROLLS (QuALITY):** 6-point accuracy improvement over Mamba-2 on multi-hop reading comprehension requiring cross-paragraph reasoning.
- **Many-shot ICL:** At 64-shot, HMM accuracy is 3 points above Mamba-2 and continues improving; Mamba-2 saturates and slightly degrades.
- **PG-19 language modeling:** 0.4 perplexity reduction at 16k tokens; 0.9 reduction at 32k tokens, showing increasing benefit at longer contexts.
- **Parameter overhead:** $<5\%$ additional parameters for the PLS extractor and LTM projections.

Ablations and Interpretation

Paragraph size P . Smaller P gives richer PLS resolution but increases LTM bank growth rate and retrieval cost. $P = 128$ – 256 gives the best quality/cost tradeoff in experiments.

LTM vs. no LTM. Removing the LTM (keeping only PLS extraction) gives intermediate improvement over base Mamba but still degrades on the longest sequences, confirming that persistent retrieval—not just better compression—is needed.

Retrieval head count. Using 4 retrieval attention heads vs. 1 improves SCROLLS accuracy by 2 points; beyond 4 heads, gains are marginal.

Reference

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